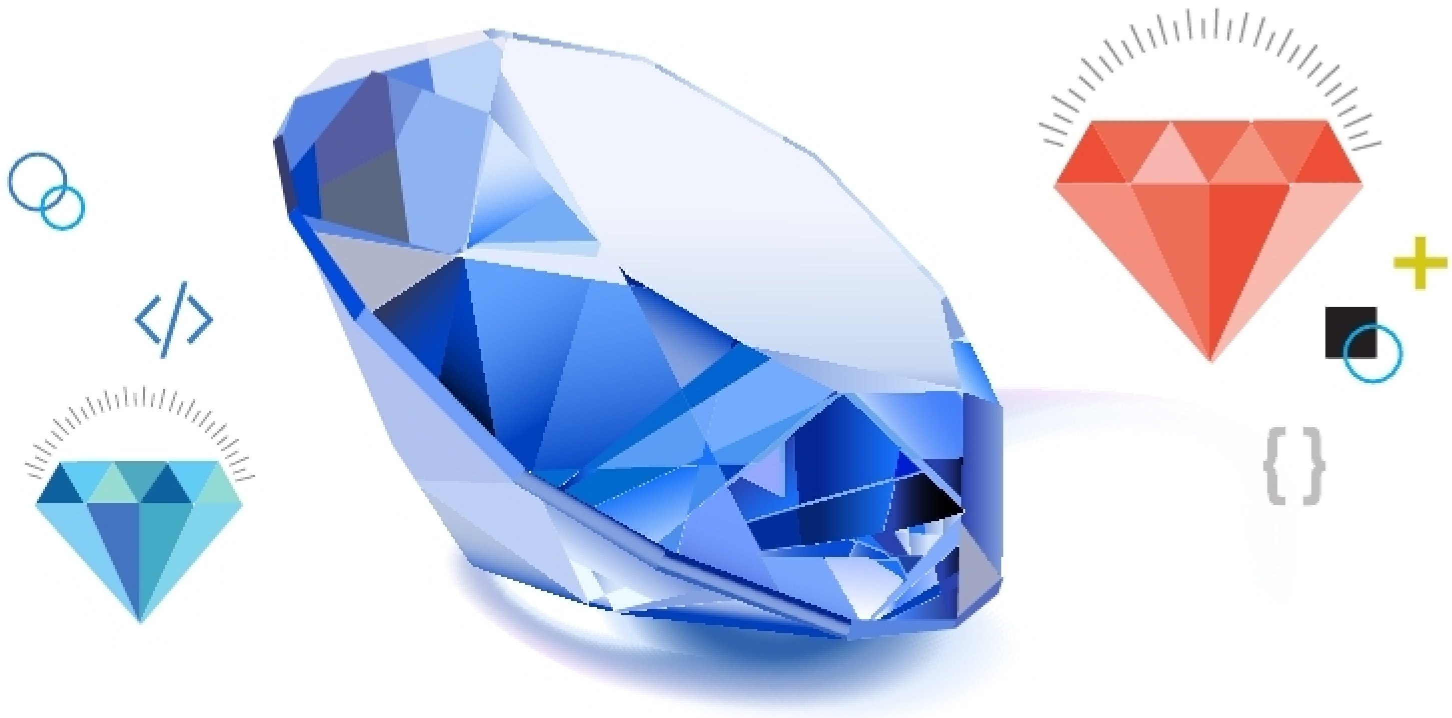
A Comprehensive Guide to Programming *in* Ruby

This article gives readers Ruby in a nutshell, covering all aspects of how to program in it. Enthusiasts can then get started on it quickly.



Ruby is a cross-platform language, and like Java, it is also an OOP (object-oriented programming) language. It is a general-purpose scripting language, which is dynamic, powerful, easy-to-use and reflective.

You can use Ruby as a flexible programming language for the following:

- Developing Web applications
- Processing text
- Developing games

With Ruby, you can perform the following tasks:

- Write Common Gateway Interface (CGI) scripts
- Embed it into Hypertext Markup Language (HTML)
- Develop Internet and intranet applications
- Install it in Windows and POSIX environments
- Connect to PostgreSQL, DB2, MySQL, Oracle, Sybase and others
- Support many GUI tools such as Tcl/Tk, GTK and OpenGL
- Make Web applications using Ruby on Rails, Sinatra or other frameworks

Quick facts

- *Developed by:* Japanese computer scientist and software programmer, Yukihiro Matsumoto (Matz), in 1993
- *Platforms supported:* Windows, Mac OS, and the various versions of *NIX
- Used by Amazon, BBC, Cisco, CNET, IBM, JP Morgan, NASA, Yahoo and many others

Rails and Rspec

Users can use the Ruby programming language to design and build higher level, domain specific languages or DSLs like Rails and Rspec.

Rails: Ruby on Rails, or just Rails, is a server-side Web application framework written in Ruby. It is released under the MIT licence. Rails is a model–view–controller (MVC) framework that provides users with default structures/scaffolds for creating Web services or pages, and even for connecting to databases.

***RSpec*:** RSpec has been written in Ruby. It is a unit test framework and is used to test the behaviour of an application. The focus is not on how the application works but on what it does.

Basic syntax

Ruby is a free format language, which means you can write code anywhere, in any line or column, without writing a main method/function.

Naming conventions

- Variable, file and directory names should be in snake case and lower case. Example: *num_name*.
- Constants should be in snake case and upper case. Example: NUMBER_OF_RECORDS.
- Ruby source code files should have an extension *.rb*. Example: *file1.rb*.
- Class and module names should be in camel case. Example: MyFirstClass.